# Likelihood fmri

## Parametrisation

This is special parameterisation of the non-central $\chi_\nu$ distribution. Let $\{x_i\}$ are iid Gaussians with mean $\mu$ and precision $\tau$, then

$$z = \sqrt{\sum_{i=1}^{\nu} \tau x_i^2}$$

is non-central $\chi$-distribution with (integer and fixed by design) $\nu > 0$ degrees of freedom, and non-centrality parameter

$$\rho = \sqrt{\nu \tau \mu^2}.$$

The observation $y$ is $y = z/\sqrt{\tau}$ and we're interested in the underlying true signal

$$\lambda = \rho/\sqrt{\tau} = \sqrt{\nu \mu^2}$$

## Link-function

The linkfunction is given as

$$\log(\lambda) = \eta$$

where $\eta$ is the linear predictor.

## Hyperparameters

The hyperparameters are $\theta = (\theta_1, \theta_2)$, where

$$\tau = \exp(\theta_1)$$

is the precision, and

$$\nu = \theta_2$$

For technical reasons, $\nu$ is implemented as a hyper-parameter, but is required to be fixed. Hence, the initial value for $\theta_2$ defines the (fixed) value for $\nu$.

The prior is given on $\theta_1$.

## Specification

- `family="fmri"` or `family="fmrisurv"`
- Required arguments: $y$ (and optional `scale` for `fmri` to scale $\tau$)

### Hyperparameter spesification and default values

**doc** `fmri distribution (special nc-chi)`

**hyper**

**theta1**

**hyperid** `103101`
**name** `precision`
**short.name** `prec`
**output.name** `Precision for fmri`
**output.name.intern** `Log precision for fmri`

**initial** 0
**fixed** FALSE
**prior** loggamma
**param** 10 10
**to.theta** function(x) log(x)
**from.theta** function(x) exp(x)

**theta2**

**hyperid** 103202
**name** dof
**short.name** df
**output.name** NOT IN USE
**output.name.intern** NOT IN USE
**initial** 4
**fixed** TRUE
**prior** normal
**param** 0 1
**to.theta** function(x) x
**from.theta** function(x) x

**survival** FALSE

**discrete** FALSE

**link** default log

**pdf** fmri

**doc** fmri distribution (special nc-chi)

**hyper**

**theta1**

**hyperid** 104101
**name** precision
**short.name** prec
**output.name** Precision for fmrisurv
**output.name.intern** Log precision for fmrisurv
**initial** 0
**fixed** FALSE
**prior** loggamma
**param** 10 10
**to.theta** function(x) log(x)
**from.theta** function(x) exp(x)

**theta2**

**hyperid** 104201
**name** dof
**short.name** df
**output.name** NOT IN USE

**output.name.intern** NOT IN USE

**initial** 4

**fixed** TRUE

**prior** normal

**param** 0 1

**to.theta** function(x) x

**from.theta** function(x) x

**survival** TRUE

**discrete** FALSE

**link** default log

**pdf** fmri

## Example

In the following example we estimate the parameters in a simulated example.

```
n <- 300
x <- rnorm(n, sd = 0.3)
df <- 1
prec <- 3
eta <- 1 + x
lambda <- exp(eta)
y <- sqrt(rchisq(n, df = df, ncp = prec * lambda^2) /prec)

r <- inla(y ~ 1 + x,
          data = data.frame(y, x),
          family = "fmri",
          control.family = list(hyper = list(df = list(initial = df))),
          control.inla = list(cmin = 0,
                              int.strategy = "eb",
                              strategy = "adaptive"),
          verbose = TRUE)
summary(r)

## 'cmin=0' seems to be required only for initial values that can give
## 'crazy' values. We can rerun without this re-starting at the prev fit,
## to validate
r$.args$control.inla$cmin <- -Inf
r$.args$control.inla$int.strategy <- "auto"
rr <- inla.rerun(r)
summary(rr)
```

## Notes

Thanks to LS for providing all the details and a robust implementation of this likelihood.